The next steps

- Today
  - Summary of model selection exercise etc.
  - Lecture on multivariate methods
  - Exercise session (multivariate methods)
- Monday
  - Lecture on causal inference
  - Intro to “midterm” exercise (practice exam)
  - Seminar on the use of simulations to test hypotheses
  - Exercise session (causal inference)
- Wednesday
  - Working on the midterm exercise
  - Seminar?

Processing and Analysis of Biological Data

BIOS15 2025

Lecture 9.
Multivariate analysis

$$y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i$$

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Univariate vs. Multivariate analyses

- So far we have worked only with one (response) variable at a time, which we refer to as univariate analyses
- (Even a multiple-regression model is called a univariate analysis)
- Biological data are often multivariate, and we are often interested in patterns of covariation among variables (e.g. phenotypic measurements or climatic variables)

$$\begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}$$

Variance matrices

- We can summarize multivariate data by symmetrical variance matrices, with variances on the diagonal and covariances on the off-diagonals

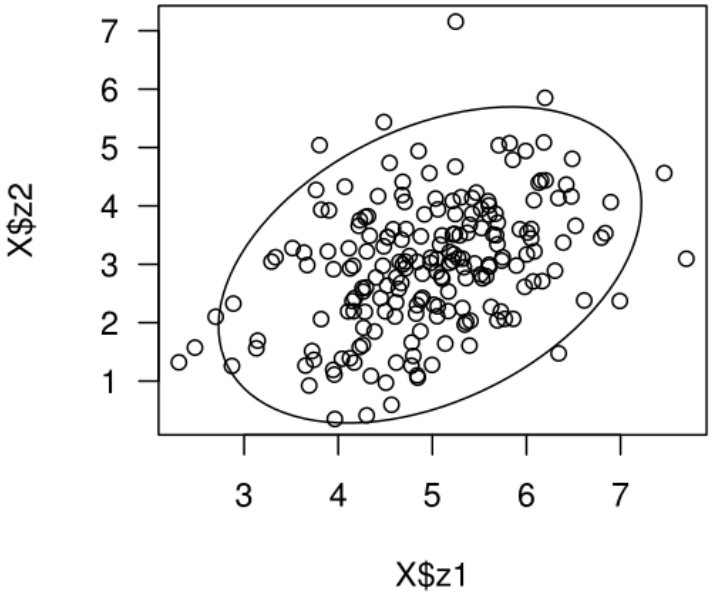
$$\begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}$$

The multivariate normal distribution

- We can summarize multivariate data by symmetrical variance matrices, with variances on the diagonal and covariances on the off-diagonals
- The multivariate normal distribution is normal in multiple directions, and is defined based on a vector of means and a variance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}$$

$$\sim MVN(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$



Eigenanalysis of variance matrices

- A vector v is an eigenvector of a matrix **A** if

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- Each eigenvector has an associated eigenvalue λ
- This yields the relation

$$\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^{-1}$$

where **Q** is a matrix with the eigenvectors in columns, and $\boldsymbol{\Lambda}$ is a diagonal matrix with the eigenvalues on the diagonal (and zeros elsewhere)

Motivating example

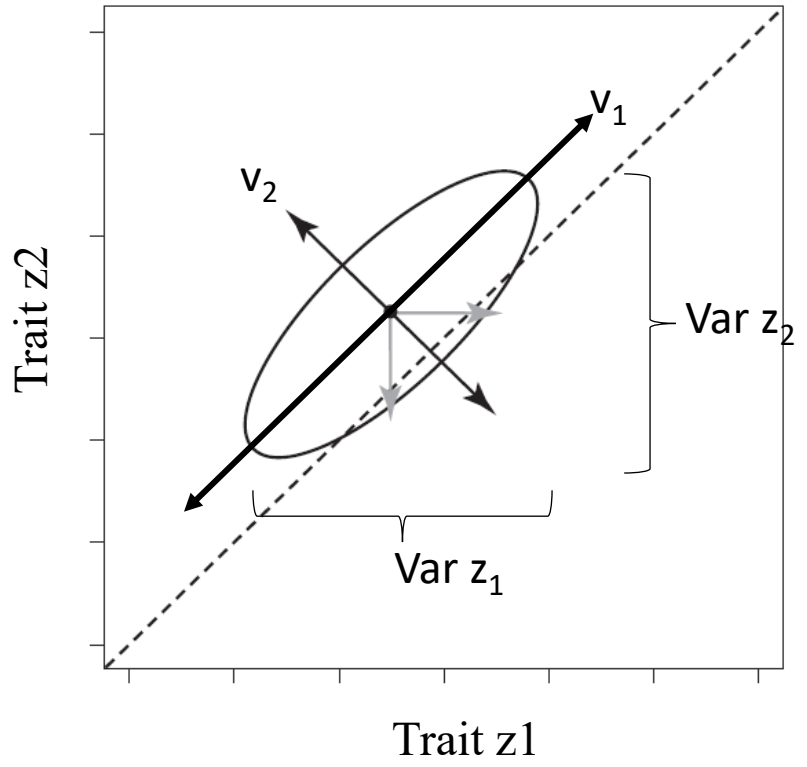
- **Leslie matrices** describe the dynamics of an age-structured population
- The first eigenvalue is the population growth rate, and its eigenvector gives the stable age distribution

$$\begin{bmatrix} n_0 \\ n_1 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_{t+1} = \begin{bmatrix} f_0 & f_1 & f_2 & \cdots & f_{\omega-2} & f_{\omega-1} \\ s_0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & s_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & s_{\omega-2} & 0 \end{bmatrix} \begin{bmatrix} n_0 \\ n_1 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_t$$

Eigenanalysis of variance matrices

- The eigen-decomposition allows us to reorganize the variance into a set of orthogonal directions so that the first eigenvector represents the direction of most variance, and the last eigenvector the direction of least variance.
- The eigenvalues represent the amount of variance associated with each eigenvector

$$\mathbf{Av} = \lambda\mathbf{v} \qquad \mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^{-1}$$



Eigenanalysis in R

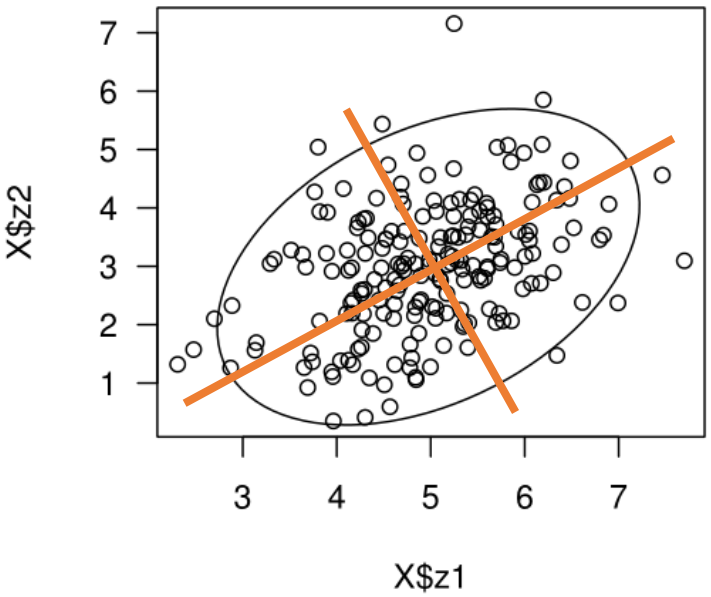
- The `eigen` function returns the eigenvectors and their corresponding eigenvalues. The eigenvectors are given in columns.

```
eigen(cm)
```

```
## eigen() decomposition
## $values
## [1] 1.4034727 0.9349665 0.1615607
##
## $vectors
##            [,1]       [,2]       [,3]
## [1,] 0.07739007  0.8285983  0.5544688
## [2,] 0.90374270  0.1765485 -0.3899741
## [3,] 0.42102245 -0.5312773  0.7351766
```

Principal component analysis (PCA)

- Principal components (PCs) correspond to the directions given by the eigenvectors
- The first (leading) PC has the largest variance, and thus represents the multivariate direction with most variance in the data
- The last (trailing) PC has the least variance

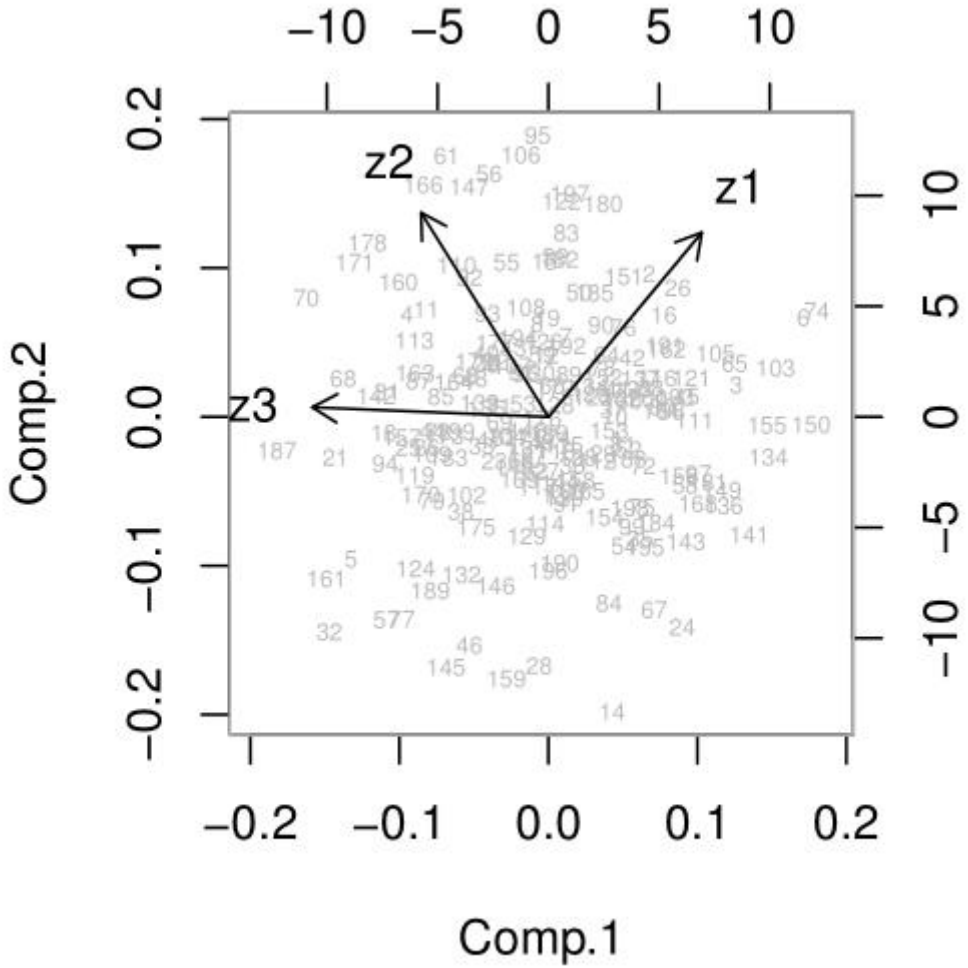


PCA loadings and biplot

- Loadings are essentially eigenvectors (though subject to some additional rotation and scaling)
- Greater values means greater "weight" of that variable on that PC

```
pca$loadings[1:3, 1:3]
```

```
##         Comp.1     Comp.2     Comp.3
## z1  0.4963374 0.66979767  0.5522864
## z2 -0.4120311 0.74171259 -0.5292379
## z3 -0.7641201 0.03512142  0.6441172
```



Principal component regression

- One application of PCA is to construct a reduced set of predictors to include in a regression model.

```
##
## Call:
## lm(formula = y ~ pc1 + pc2 + pc3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4644 -0.6514 -0.0769  0.6864  3.7741
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.09550    0.07499   1.273   0.2043
## pc1          0.13296    0.06280   2.117   0.0355 *
## pc2          0.31012    0.06420   4.830 2.74e-06 ***
## pc3          0.88360    0.16988   5.201 4.96e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.061 on 196 degrees of freedom
## Multiple R-squared:  0.2187, Adjusted R-squared:  0.2067
## F-statistic: 18.29 on 3 and 196 DF,  p-value: 1.672e-10
```

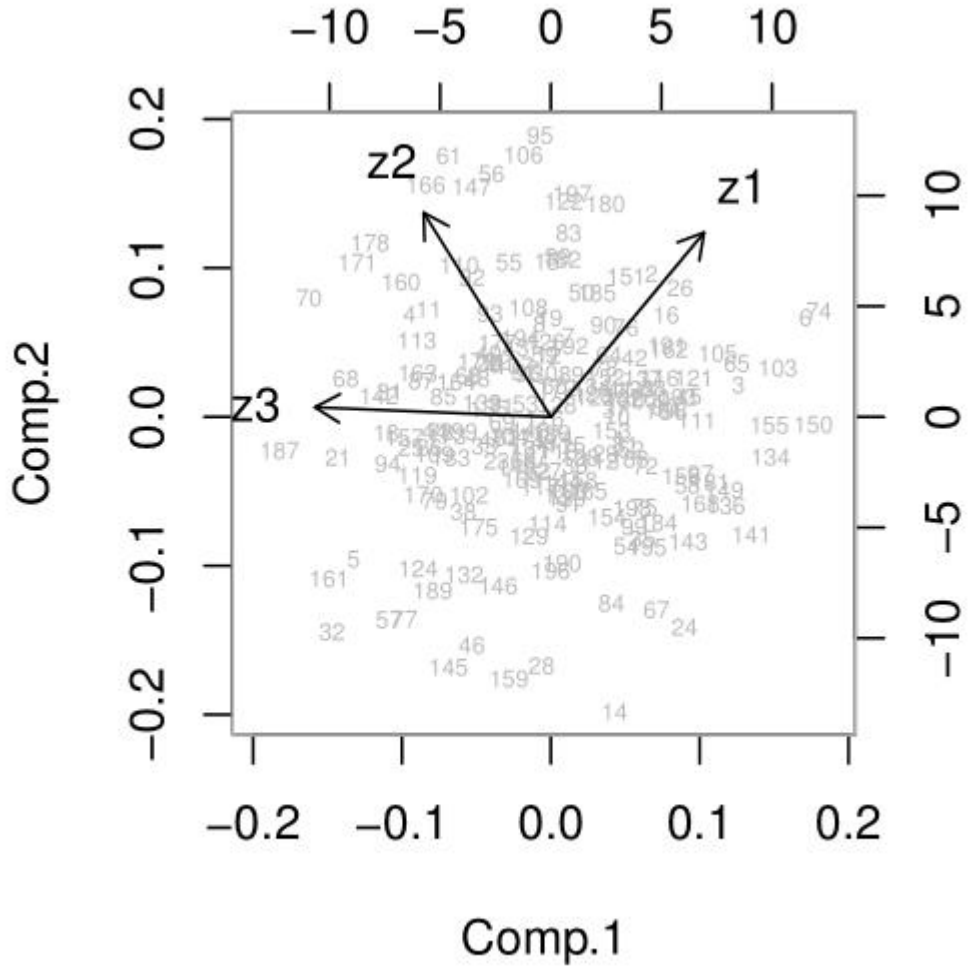
Principal components do not add anything, they are just a different formulation of the same variance

```
##
## Call:
## lm(formula = y ~ XX$z1 + XX$z2 + XX$z3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99179 -0.81727 -0.02969  0.82259  2.67106
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09966    0.07748  -1.286   0.1999
## XX$z1        0.45250    0.10221   4.427 1.58e-05 ***
## XX$z2       -0.19661    0.10028  -1.961   0.0513 .
## XX$z3        0.16684    0.11032   1.512   0.1320
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.096 on 196 degrees of freedom
## Multiple R-squared:  0.1024, Adjusted R-squared:  0.08868
## F-statistic: 7.455 on 3 and 196 DF,  p-value: 9.423e-05
```

```
##
## Call:
## lm(formula = y ~ pc1 + pc2 + pc3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99179 -0.81727 -0.02969  0.82259  2.67106
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09966    0.07748  -1.286  0.19988
## pc1          0.17812    0.06281   2.836  0.00505 **
## pc2          0.16311    0.07043   2.316  0.02160 *
## pc3          0.46143    0.15417   2.993  0.00312 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.096 on 196 degrees of freedom
## Multiple R-squared:  0.1024, Adjusted R-squared:  0.08868
## F-statistic: 7.455 on 3 and 196 DF,  p-value: 9.423e-05
```

Dimension reduction through principal-component regression

```
##
## Call:
## lm(formula = y ~ pc1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.39164 -0.82388  0.01201  0.83536  2.72336
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.06918    0.07749  -0.893   0.3731
## pc1          0.15401    0.06222   2.475   0.0142 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.096 on 198 degrees of freedom
## Multiple R-squared:  0.03001,    Adjusted R-squared:  0.02511
## F-statistic: 6.126 on 1 and 198 DF,  p-value: 0.01416
```



Writing (analysis) methods

- Normally a specific section at the end of the Methods (but sometimes integrated throughout)
- Focus on the aim of the analysis before technical details
- For models, list terms in words, sometimes include model equation
- R syntax is increasingly acceptable

Examples of model description

We analysed seed set per blossom by fitting linear mixed effect models with timing of pollination, pollen type and population, as well as their interactions, as fixed factors, and maternal individual as a random factor. Although seed number is a count variable, the distribution of residuals was close to normal, allowing for the use of a linear model. To analyse variation in seed mass, we fitted linear mixed effect models with timing of pollination, pollen type and population, as well as their interactions, as fixed factors. We further included seed set and peduncle diameter as covariates to account for a possible trade-off between seed mass and seed number within seed set, and possible blossom size effects, respectively. Blossom identity nested

To investigate the relationship between microclimate and plot richness, we fitted a mixed-effects Poisson regression model with species richness of the sample plots as the dependent variable, and microclimate variables as possible explanatory variables. To account for the structure of the data (sites nested within areas), site and area were entered as random factors. At the among-site scale (using each site

bruster et al. 2009b). We modeled pollen arrival during the female phase, $P_F(z')$, in units of pollen grains, as a function of bract area, gland area, and gland-stigma distance through a log link with a negative-binomial error distribution. The minimal model describing the predicted pollen load on the last day of the female phase was

$$P_F(z') = e^{a_2 + b_{21} UBA + b_{22} GA + b_{23} GSD}. \quad (2)$$

Writing results

- Start with summary statistics/patterns of variation

Field study

RESULTS

Phenotypic traits

In the field, average ASD varied nearly threefold among populations (range = 2.35–6.32 mm, Table S2), with 48.6% of the variance due to differences among populations (Table S4). Average gland size (17.43–34.62 mm^2) and average bract size (254.96–526.32 mm^2) varied approximately twofold among populations (Table S2), with differences among populations accounting for 28.1 and 19.9% of the variance respectively.

Meta-analysis/review

Database description

The updated database (Table S3) contains 792 evolvability estimates for 54 taxa representing 27 families. Among the 72 studies included, 68.6% were conducted on populations originating in North America (Fig. S1). Most studies were conducted in glasshouses or other controlled environments.

Writing results

- Start with summary statistics/patterns of variation
- Focus on biology over statistics (explain the result in biological terms)
- Quantify and exemplify (*y* increased by 0.5 mm per mm increase in *x*)

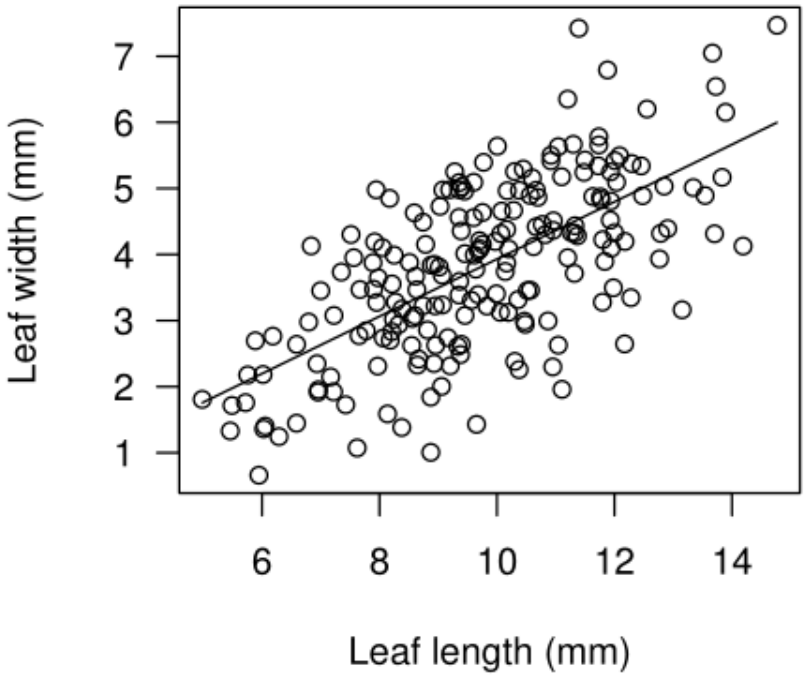
Overall, blossom development tended to be more rapid under dry conditions. In the wet treatment, the total receptive period of the blossoms (from the first day of opening to the abscision of the male cymule) lasted for 6.87 ± 0.09 d in the two large-glanded populations and 5.80 ± 0.10 d in the two small-glanded populations (fig. 4). In the dry treatment, the length of this period was moderately reduced by, on average, 5.4% across all four populations (fig. 4; table 4; see table A3 for model-selection results).

Writing results

- Start with summary statistics/patterns of variation
- Focus on biology over statistics (explain the result in biological terms)
- Quantify and exemplify (*y* increased by 0.5 mm per mm increase in *x*)
- No discussion (normally)
- Avoid unnecessary introductory sentences like “Population means are reported in Table 1”.
- Refer to tables and figures after a statement (“Body size varied substantially among populations (range = X.X to Y.Y, Table 1)”.

Making nice figures

- Labels and units!
- Keep regression lines within the data range
- If several symbols/colors: legend inside the graph
- Usually quite square panels look the best (not very rectangular)



There are also other packages for graphics

ggPlot



base R



Making nice tables

- No vertical lines
- All parameters explained in legend

Table 5

Inaccuracy Statistics under Wet and Dry Experimental Conditions

Taxon, population, treatment	Bias²	Male variance	Female variance	Joint inaccuracy (95% CI)	Mean-scaled inaccuracy (95% CI)
Large-glanded species:					
Tulum:					
Wet	.14 (16.1%)	.25 (28.8%)	.48 (55.1%)	.87 (.62, 1.17)	.03 (.02, .04)
Dry	.05 (3.0%)	1.00 (53.8%)	.80 (43.2%)	1.85 (1.42, 2.33)	.11 (.08, .14)
Puerto Morelos:					
Wet	.53 (25.8%)	.94 (46.0%)	.58 (28.2%)	2.05 (1.56, 2.66)	.07 (.05, .09)
Dry	1.42 (56.7%)	.69 (27.4%)	.40 (15.9%)	2.51 (1.94, 3.09)	.14 (.11, .17)
Small-glanded species:					
Cozumel:					
Wet	1.83 (63.7%)	.15 (5.1%)	.9 (31.3%)	2.87 (2.10, 3.71)	.22 (.17, .28)
Dry	.93 (50.2%)	.19 (10.5%)	.73 (39.4%)	1.85 (1.21, 2.57)	.20 (.14, .27)
Valladolid:					
Wet	.55 (38.8%)	.33 (23.5%)	.53 (37.7%)	1.41 (.82, 2.20)	.12 (.07, .19)
Dry	.29 (37.7%)	.24 (30.9%)	.24 (31.4%)	.77 (.43, 1.23)	.08 (.04, .13)

Note. Bias² is the mean squared deviation from the hypothesized adaptive optimum. The reported percentages are the proportion of the joint inaccuracy explained by each component. To obtain the mean-scaled inaccuracy, the joint inaccuracy was scaled by the product of the male and female trait means. Ninety-five percent confidence intervals (95% CIs) were obtained from 1000 nonparametric bootstrap estimates of the joint and mean-scaled inaccuracies, respectively.